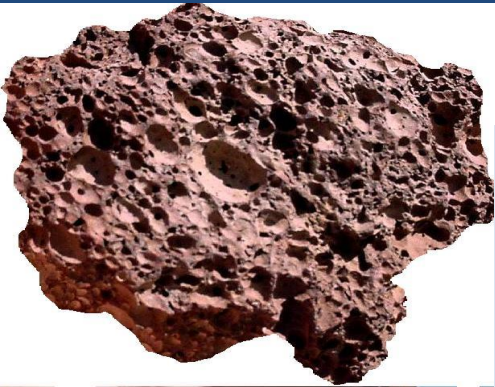


A Pre-Feasibility study to establish a quarry to extract volcanic tuff ores



2017



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1. Executive Summery

The project aims to establish a quarry to exploit and treat the volcanic tuff ore to be marketed to the domestic community, as well as, for export.

The volcanic tuff ore is exists in several areas of the kingdo, and at Tafilah in particular with huge quantities able to be exploited to increase the production to supply the local market of this important ore, which has many uses, most important is the agricultural uses.

The project will generate 11 job opportunity in case of execution.

Based on the market share and prices which was considered for the proposed project, the estimated revenues will be 1,210,000 JD during the first year of the project life, increased gradually to 1,578,776 JD during the 10th year, the payback period is 3 years, and the internal rate of return will be 36.19%. The following table shows the study results.

Table1 : Economic Indicators

Indicator	
Total Investment Cost (Jordan Dinars)	1,024,445
Internal Rate of Return (IRR)	%36.19
Pay-Back Period (Years)	3
Net Present Value (Jordan Dinars)	2,281,213
Job Opportunities provided by the Project	11
Total Revenues (For 10 Years)	13,871,294
Total Operating Expenses (For 10 years)	8,315,841
Total Net Profit (For 10 years)	2,491,477

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

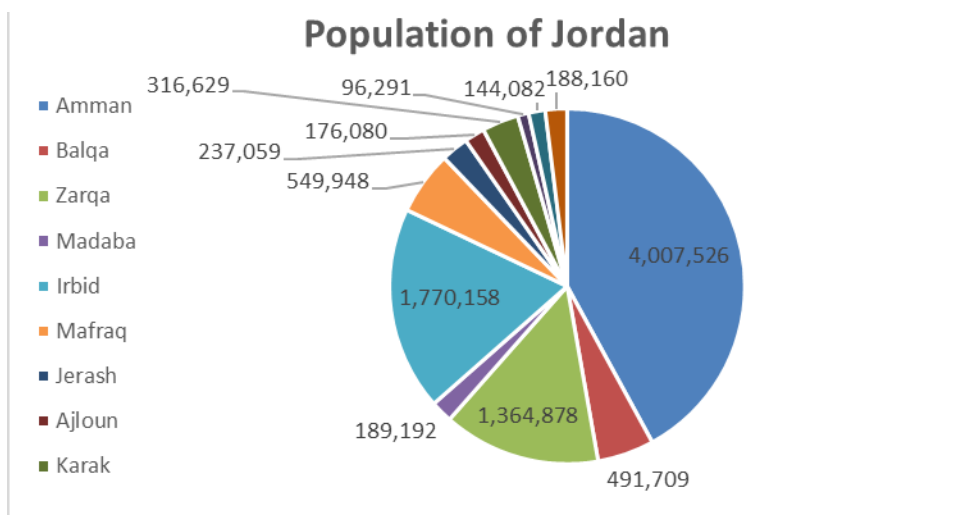
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1: Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 :Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

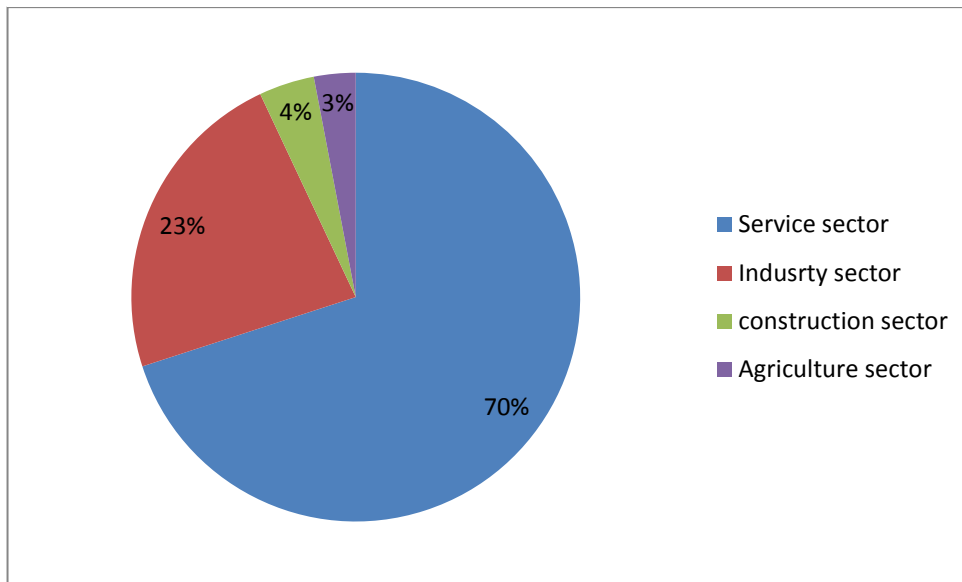
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2: Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance

of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment

Investment Climate in Al Tafilah:

Mining and Mineral Resources Sector:

Al Tafilah Governorate contains various mineral and industrial rocks, which make large Jordanian companies, decide to invest in the Governorate although a large reserve of mineral resources are still not utilized.

Jordan Phosphate Mines Company extracts phosphate exist in Al Hassa Area and the total production of phosphate fertilizers reached 5.4 Million Tons in 2013, Jordan Cement company utilizes raw materials to produce cement in Al Rashadyeh factory as the main resources are limestone and calys.

One of the main industrial rocks exist in the Governorate is the Oil shale in Jurf Al Draweesh area, Perlite near Al Hassa area, Dolomite near Al Rashadyeh area, Gypsum in Maleeh Mountain besides Al Tafilah, Feldspar and Phosphate rocks.

One of the main minerals exist in the Governorate is the Copper, as the total estimated reserves of the pure mineral is one Million Tons Cubic Meter, where total estimated reserves of the manganese is half Million Ton Cubic Meter. Source: Investment Commission, 2015.

One of the main Investment Opportunities exist in Al Tafilah Governorate is:

- Utilizing Uranium materials for producing nuclear energy
- Construction Materials Industries (Gypsum, Zeolite)
- Chemical Industries and Fertilizers production
- Utilizing Oil Shales for producing energy

3. Market analysis

Project description

The establishment of a quarry to extract the volcanic tuff at a production capacity of about 100,000 ton/year, to get a very fine sizes used in different industries, to be followed by packaging and marketing process.

The volcanic tuff is one of the pyroclastic rocks occurred by precipitation and aggregation of hard and violated particles where the volcanoes are pushing it to far distances and near to the nozzle of the volcanoes. Based on volume and weights of these particles, its can be divided into three types as shown in table 3 below:

Table 3: types and sizes of volcanic tuff

Type	Size (mm)	Shape
Agglomerate	32	Rounded or sub rounded
Volcanic Brescia	32	Angular shape
Volcanic Tuff	4-32	Different size and shape

Ersen book 1976

The rock can be divided into two main groups:

1. An Agglomerate and the Volcanic Brescia which been formed after its precipitated as volcanic rocks which produced of rough flowed Agglomerate with rounded shapes, the volcanic Brescia is medium sizes flowed that have sharp angles with size 32 mm.
2. Volcanic tuff is formed from aggregates, ash and volcanic dust the sizes are ranged between 4-32 mm for the volcanic aggregates and less than 4 mm for ash and volcanic dust.

Types of volcanic tuff:

The volcanic tuff is classified according to the physical and chemical properties, (Size and type of grains) as follows:

Chemical properties:

The volcanic tuff is divided based on its chemical structure to two type:

- Acidic Tuff

The acidic tuff contain a high percentage of Silica 66%, at the same time contains a high percentage of (Al, K, Na) and these type contains a low percentage of (Mg, Ca, Fe) for this reason it contain of high percentage of white minerals and brightness color.

- Basic Tuff

In this type the silica content is less than the acidic tuff about 45% and this type is rich with black minerals as a result of high percentage of (Ca, Mg, Fe).

Physical properties

- **Particles size**

The volcanic aggregate Lapilli is medium and fine particles tuff ranged between 4-32mm contains of hard lapilli with different colors and sizes its exists in different colors: yellow, gray, brown, the yellow tuff is considered as the bestone due to its hydraulic properties.

An ash and volcanic dust: which is tuff with very small particles size as of volcanic dust and formed after precipitation in far areas of volcanoes, the volcanic dust or ash which is the best types of the volcanic tuff with size ranges 0.01-4mm.

Particular Types:

Vitric tuff: formed mainly from volcanic glass pieces with particle sizes less than 4 mm, angular, and lighted brown to dark color.

Crystalline tuff: its formed from pieces of crystals shapes.

Lithic tuff: This formed from hard or cohesive rock pieces with greater than 4mm particle size.

- Hydraulic properties: hydraulic factor is ranged between 26-72 HI, and this factor is high in the fine particles.
- Specific gravity: average of specific gravity is about 1.6 g/cm³.
- Colors of the tuff: it have different color such as (red, yellow, gray, black), the yellow tuff is the best type.

The chemical analyses for the volcanic tuff have proved that its basic tuff type of basaltic origin, as shown in the table 4 below:

Table 4:the chemical analysis for the tuff

Element	Percentage
SiO ₂	40-45%
FeO	10.5-12.5 %
MgO	5-10%
Alumina	10.5-12.5 %

Source: Natural resources authority

Mineralogy of volcanic tuff:

Major minerals: Blygoclase, Nafline, Olivene, pyroxene, magnetite

Minor minerals: calcite, hematite, zeolite

The geological studies indicated that the volcanic activity in Jordan is started since 28 million year, continues as intervals for half million year, and the volcanic tuff is spread in Jordan in three major locations:

- The northern area: Azraq area
- The middle area: deposits of Zara, Wadi Hidan and Shehan deposit.
- The southern area: Jurf Adaraweesh, Alshobak and Tafilah.

Project Objectives

Quarry to Extract Volcanic Tuff Ores- Tafilah

- Exploiting volcanic tuff ores, and encouraging investors to enter into the fields of manufacturing.
- Increase the export to region and international markets.
- Producing materials with high quality and competitiveness to help in developing the local community.
- To generate a new jobs to the local community.

The Proposed product

The most important use of volcanic tuff is the agricultural use:

- Its added to organic fertilizers
- Used to keep the humidity of soil, and absorption the vibrated water in the air
- Fertilizer and soil enhancement.
- Used to remove the ammonia from the agriculture and industrial waste

The volcanic tuff is used in many construction industries and products such as:

- Insulator materials industry.
- Light concrete industry.
- Cement board industry.

Targeted categories

The project targeted many categories such as:

Agricultural Nurseries

Decoration and building companies such as:

- Aqmesha Company for decorations
- World of Decoration Company
- Engineering company for decorations

Cement factories such as:

- Lafarge factory
- Almanaseer factory
- White cement factory

- others

Maket analysis

An estimated reserves of Jordan volcanic tuff is about 1340 million metric ton (natural resources authority), the annual industrial usage in cement manufacturing is about 412,000 ton, and the reserves of the southern regions is about 500 million ton as shown in the table 5 below:

Table 5:the volcanic tuff reserve in southern region

Region	The reserve (million ton)	Ore thickness
AlShoubak	153	38
Jurf AlDaraweesh	124.8	34
Tafilah	160	25
Jabal Enesa	62.4	50

Source: Natural resources authority

Competitors analysis

The important locations where the volcanic tuff is extracted are Tull Remah, Tull Husban which are located at the northern eastern region of the kingdom.

The Jordanian cement Factories at Alfuhes was using the local production of volcanic tuff in cement industry (Portland cement and bozzalana), the total amounts used is about 412,000 ton/year with a cost of 2 million JD; the cost of one ton 4,375 JD FOB Factory.

The most important companies producing volcanic tuff ores:

- Emar Company
- Amanah Company for agriculture and industrial investments
- Jordanian company for soil enhancement
- Aljaleel Company for mining
- Modern Company for cement and mining

Table6 :An annual production of some volcanic tuff companies.

Company/ year	2008	2010	2012	2014	2016
Jordanian cement factories	224788	103991	153529	248144	250156
Modern company for cement and mining	The company obtained on mining right in 2014				180616
Amana company for agriculture and industrial investment	608	63205	0	14292	16965
Jordanian company for soil enhancement	1218	2751	0	0	6458
Aljaleel minig company	6970	308718	118208	212995	225489

Source: Natural resources authority.

Market Share

Jordan exports of volcanic tuff in 2014 was about 19,297 ton, and the local production of tuff is about 1,051,264 ton/year, based on that the quarry will be established for extracting and mineral beneficiating of volcanic tuff ores with 0.5% market share and annual production of 50,000 ton/year; 166 ton/day and 8 hours/ day.

Table 7: The Expected Market Share

Type	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Producing volcanic tuff percentage										
Agriculture decoration in usage	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%
Producing volcanic tuff percentage										
Water purification usage	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
Producing volcanic tuff percentage										
Agriculture	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%

Prices analysis and pricing policy

The prices of volcanic tuff ores are ranged between 8-50 JD based on particle size distribution and industrial uses, the production capacity of the quarry is about 50,000 ton/year, three types of volcanic tuff with various uses will be produced as shown in table 8 below:

Table8 :Price Analysis

The use	Percentage %	Production (Ton)	Price (JD)	The revenue
Agriculture decoration	40	20,000	8	160,000
Water treatment	30	15,000	50	750,000
For Agriculture	30	15,000	20	300,000
Total .	100	50,000	---	1,210,000

Expected Revenues Analysis

The following table shows the annual exoected revenues for ten years(2018-2027):

Table 9: Annual Expected Revenues for 10 years

Prices Assumptions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
--------------------	------	------	------	------	------	------	------	------	------	------

Selling price of 1 ton volcanic tuff agriculture decoration usage	8	8	8	9	9	9	10	10	10	10
Selling price of 1 ton volcanic tuff water purification usage	50	52	53	55	56	58	60	61	63	65
Selling price of 1 ton volcanic tuff for agriculture	20	21	21	22	23	23	24	25	25	26
Production ratio of volcanic tuff agriculture decoration usage	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%
Production ratio of volcanic tuff water purification usage	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
Production ratio of volcanic tuff for agriculture	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
Total Revenues	1210000	1246300	1283689	1322200	1361866	1402722	1444803	1488147	1532792	1578776

4. Technical analysis

The human resources required

Table10 :The Required Human Resources

Position	No.
Project manager	1
Accountant	1
Mining Technician	1
Worker	2
Driver	3
Guard man	2
Cleaner Labour	1
Total	10

Job Description:

- General manger: Propose the general policy, preparing the plans and supervising its execution, administrative and technical supervision of the staff, development the criteria of insurance and the safety.
- Accountant: responsible for financial aspects in addition to the preparing the annual financial report.
- Technician Mining: analyse and studying the ore reserves, prices, costs and production rate.
- Labor: prospect and transport the minerals.
- Cleaning labour: keeps the working area clean.
- Security guard: secure the location of work.

Location analysis

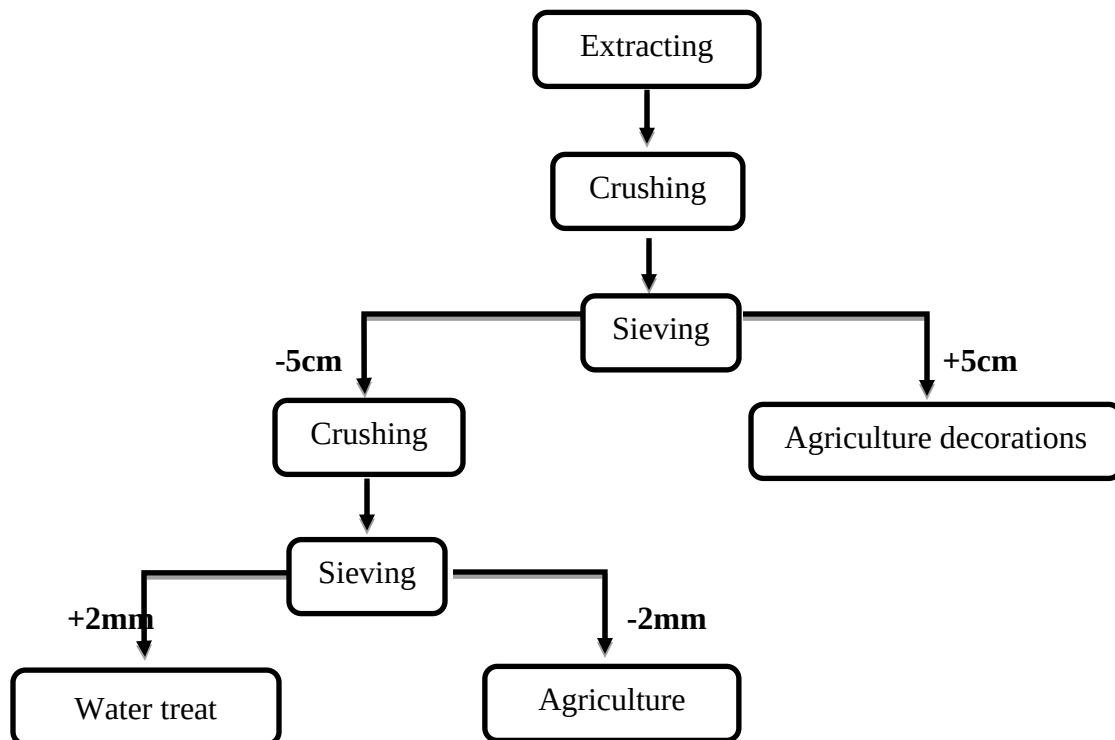
Referring to occurrences of volcanic tuff ores and reserves for each area mentioned in table3 above, the perfect choice is Jurf Adaraweesh because its closer to highway road, the reserve of this area is about 124.8 million ton. Where the ores outcrops at surface in this region, the Map.1 below shows the volcanic tuff deposits in Jordan:

Figure 3: the volcanic tuff deposit in Jordan



Description of volcanic tuff production

The extraction process for the volcanic tuff ores that outcropped at Jurf Adaraweesh region, carried out by crushing the ores into two stages: primary crushing using Jaw crusher, after that the sieving will begin to obtain two particle size (+5 cm) then the (-5 cm) crushing by Jaw crusher, finally the product will be sieved to produce particles with (+2 mm), as shown in the flow sheet below:



Technical analysis

The land and construction works

It was assumed that a land 50 dunum in Jurf Adaraweesh region will be leased after prospecting about 1km², table11 below shows the fees of mining - based on the recommendations of the

Central Committee of State property / source: organization of the energy and minerals regulatory commission:

Table11 :the fees of mining

The fees	Mining fees (JD)
Fees of prospecting and exploration license	200
Fees of annual mining lease of 1km ²	500
Fees of prospecting license granting	500
Fees of rent for prospecting aspects 1 km ²	800
Fees of annual rent is 25 JD for one dunum, 50 dunum	1,250
Total.	3,250

Table12 :construction works

Construction works	Area(m ²)
Construction building	150
Hunger building	200
Another works	---

Equipments and machines

Table 13 below shown the equipments and machines required for the project.

Table13 :Machinery and Equipment

Machine	No.
Loader	1
Secondary crusher	1
Primary crusher	1
Conique crushed	1

Sieving machine

2

The furnitures

The project is required to the following furniture:

Office furnitures (computers, tables and chairs, carpet, meeting requirements) the cost estimated at 12,000JD.

Vehicles and transportation means

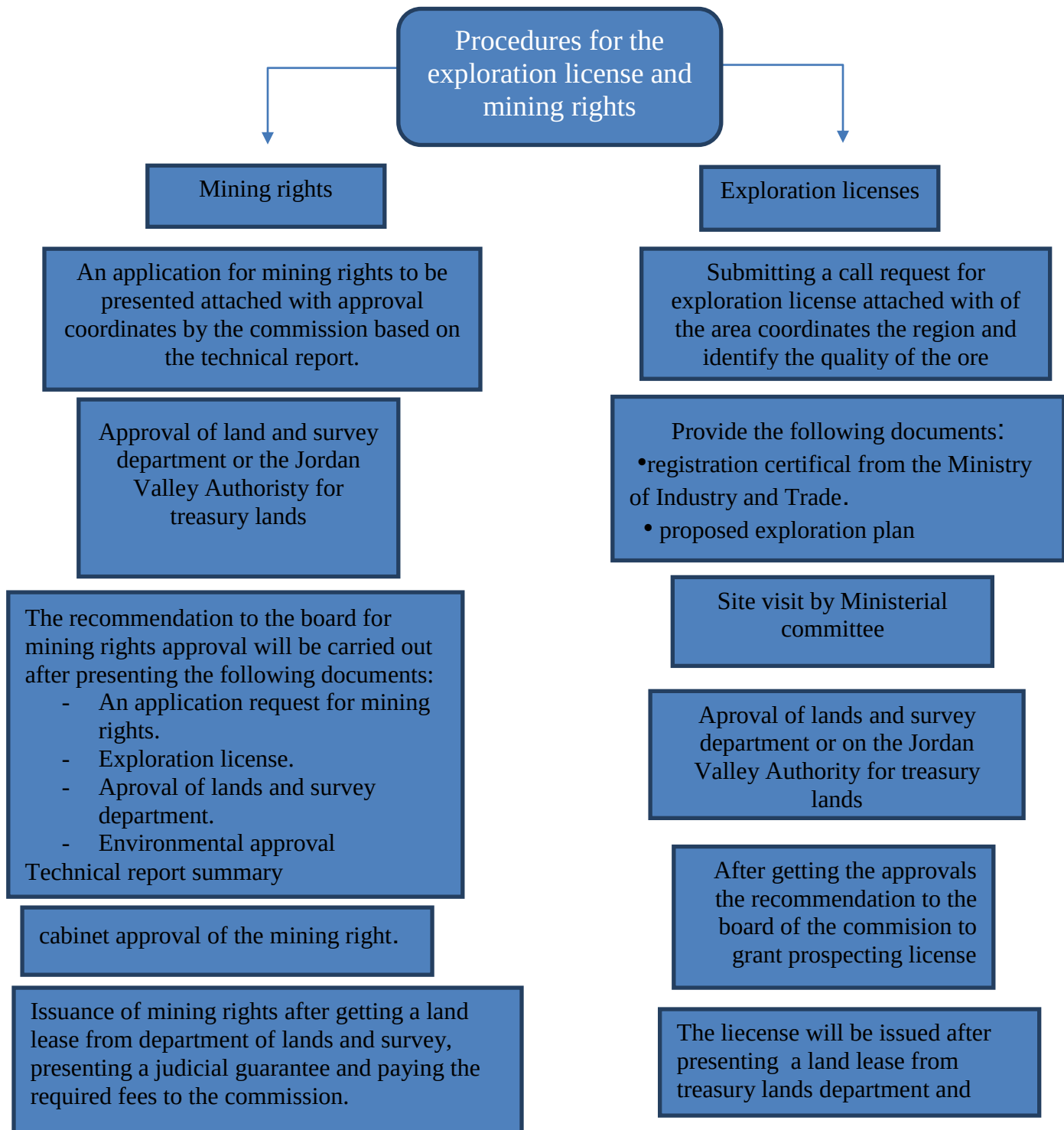
The following table shows the vehicles and transportation required for the project

Table14 :vehicles required for the project.

Vehicles and transportation means	No.
Bik up	1
Dumper	1

Legal procedure and requirements

The following figure shows the necessary procedures for exploration and mining rights license based on regularity of energy and minerals sectors commission:



5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 10.75% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 20 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Table 15:General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 16:Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 17: Project working times

Project working times	
Number of weeks/Year	52
Working Days/Week	7
Number of days / Year	300
Number of working hours/Day	8

Table 18: Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate (1)	3.00%
Annual Expenses Increase Rate (2)	3.00%
Annual Increase Rate in Employees Salaries (4)	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 19: Assumptions of expenses

Assumptions of expenses	
Expenses of raw materials of income (food and drink)	50.00%
Service benefit expenses from revenues	2.5%
Maintenance expenses from revenues	1.00%
Consumer expenses from revenues	1.00%
Insurance expenses from the value of assets	0.75%
Marketing expenses of revenue	0.25%

Table 20: Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
(2) Income Tax Deduction	0.50
Income Tax after Deduction	20%

(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 21: Risk Premium

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	20.00%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 22: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%20
Bets	0.89
Equity	409,778
Loans Value	614,667
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	614,667
Loans Percentage	%60
Equity	
Cost of Equity	%15.5

Equity Value	409,778
Equity Percentage	%40
Gross	
Project Value	1,024,445
Project Percentage	%100
Weighted Average Cost of Capital	%10.75

Capital Expenditures

The estimated cost of the project is 1,024,445 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses, working capital and vehicle requirement. The following tables summarize the details of these expenses:

Table 23: Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost/ JD
Building Construction	200	150	30,000
Hanger Works	100	500	50,000
Other Works			8,000
Total			88,000

Table 24: Machines and equipment Capital Expenditure

Description	Number	Unit Cost / JD	Gross Cost/ JD
Loader	1	60,000	60,000
Primary crusher	1	40,000	40,000
Secondary crusher	1	40,000	40,000
Sieving machine	1	40,000	40,000
Dumper	2	25,000	50,000

Total	1	60,000	60,000
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Table 25: Furniture Capital Expenditure

Description	Number	Unit Cost / JD	Gross Cost/ JD
Furniture			5,000
Total			5,000

Table 26: Transportation and Vehicles Capex

Transportation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Pick up	1	23,000	23,000
tipper	1	377,000	377,000
Total			50,000

Pre Operating Expenses

Consist of a set of expenses that are expended before the start of the project and before any revenue is realized. It consist of Government fees, licenses, marketing expenses, training and transportation expenses which are estimated as follows:

Table 27: Pre-operating expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	500
Transportation	750
Marketing	500
Training	750
Total	2,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 28: Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	181,349

General and administrative expenses	3	%25	4,349
Marketing expenses	3	%25	756
Indirect Salaries	3	%25	19,360
Total			205,814

Table 29:Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Construction works	88,000	10%	96,800	%9.4
Machinery & Equipment	230,000	10%	253,000	%24.7
Furniture	5,000	10%	5,500	%0.5
Vehicles	400,000	10%	440,000	%43.0
Pre-Operating Expenses	2,500	10%	2,750	%0.3
Working capital	205,814	10%	226,395	%22.1
Total (JD)	931,314		1,024,445	100%

Source of funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,024,445) JD, where 60% of the project will be financed through loans (614,667) JD, and the remaining (40%) through (Equity) with (409,778) JD. The following table summarizes the general structure of the required financing:

Table 30:Source of funding

Funding Sources	Amount	Percentage
Equity	409,778	%40
Loans	614,667	%60
Total	1,024,445	%100
Use of Fund:	Value	Ratio
Capital Expenditures	795,300	%77.63
Pre-Operating Expenses	2,750	%0.27
Working Capital	226,395	%22.10
Total	1,024,445	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 31: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from	605,000	623,150	641,845	661,100	680,933	701,361	722,402	744,074	766,396	789,388
Utilities from	30,250	31,158	32,092	33,055	34,047	35,068	36,120	37,204	38,320	39,469
Maintenance from	12,100	12,463	12,837	13,222	13,619	14,027	14,448	14,881	15,328	15,788
Disposables from	12,100	12,463	12,837	13,222	13,619	14,027	14,448	14,881	15,328	15,788
Others	65,945	67,923	69,961	72,060	74,222	76,448	78,742	81,104	83,537	86,043
Total	725,395	747,157	769,572	792,659	816,438	840,932	866,160	892,144	918,909	946,476

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 32: General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	600	618	637	656	675	696	716	738	760	783
Stationary and Printing	600	618	637	656	675	696	716	738	760	783
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Renting	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Training	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Insurance	5,965	6,144	6,328	6,518	6,713	6,915	7,122	7,336	7,556	7,783
Miscellaneous	1,581	1,629	1,678	1,728	1,780	1,833	1,888	1,945	2,003	2,063
Total	17,396	17,918	18,456	19,009	19,580	20,167	20,772	21,395	22,037	22,698

Marketing Expenses

The Following table shows the marketing expenses for the proposed project, which is 50% of the revenues based on the assumption of the study team:

Table 33:Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	3,025	3,116	3,209	3,305	3,405	3,507	3,612	3,720	3,832	3,947

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summarizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 34:Number of Employees

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	1	1	1	1	1	1	1	1	1	1
Accountant	1	1	1	1	1	1	1	1	1	1
Technical mining	1	1	1	1	1	1	1	1	1	1
Worker	2	2	2	2	2	2	2	2	2	2
Cleaner man	1	1	1	1	1	1	1	1	1	1
Driver	3	3	3	3	3	3	3	3	3	3
Guard man	2	2	2	2	2	2	2	2	2	2
Total Employees	11	11	11	11	11	11	11	11	11	11

Table 35: Expected Monthly salaries and wages

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Accountant	400	416	433	451	470	489	509	530	552	574
Technical mining	350	364	379	395	411	428	445	464	483	502
Worker	300	312	325	338	352	367	382	397	414	431
Cleaner man	250	260	271	282	294	306	318	331	345	359
Driver	380	396	412	429	446	465	484	503	524	546
Guard man	300	312	325	338	352	367	382	397	414	431

Table 36: Expected Annual Salaries

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Accountant	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Technical mining	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Worker	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Cleaner man	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Driver	18,240	18,988	19,766	20,577	21,420	22,299	23,213	24,165	25,155	26,187
Guard man	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Total Salaries	77,440	80,615	83,920	87,361	90,943	94,671	98,553	102,594	106,800	111,179

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 37: The following tables illustrate cost of Capex and Depreciation rates:

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions
Construction Works	96,800	%5.0	%0.0
Machineries	253,000	%10.0	%2.0
Furniture	5,500	%10.0	%5.0
Vehicles	440,000	%10.0	%0.0
Total	795,300		

Table 38: Depreciations and Additions on Construction Works

Capex, Annual Additions and	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	795,30	800,63	806,08	811,65	817,34	823,15	829,09	835,15	841,35	847,69
Total Depreciation Expenses	74,690	75,224	75,768	76,325	76,894	77,475	78,069	78,676	79,296	79,929
Total Accumulated Depreciation	74,690	149,91	225,68	302,00	378,90	456,37	534,44	613,12	692,41	772,34
Total Additions	0	5,335	5,450	5,568	5,688	5,811	5,938	6,067	6,199	6,335
Total Net Book Values	720,61	650,72	580,40	509,64	438,43	366,77	294,64	222,03	148,93	75,345

Loan

The table below summarizes all details related to the loan such as annual instalments and payment methods for the remaining amount of the loan for each year of the project to 2020.

Table 39: Loan Details

Loan Value	614,667
Annual Interest Rate	9.48%
Loans Period	5
Grace Period	0
Loan Starts at year:	2018
Annual Payment	159,999
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				614,667
2019	159,999	58,270	101,729	512,938

2020	159,999	48,627	111,373	401,566
2021	159,999	38,068	121,931	279,635
2022	159,999	26,509	133,490	146,145
2023	159,999	13,855	146,145	0

Income Statement

Based on financial information and analysis of the financial expenses of the project, the estimated income statement for the years (2018-2027) will be as shown in the following table:

Table 40: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	1,210,00	1,246,30	1,283,68	1,322,20	1,361,86	1,402,72	1,444,80	1,488,14	1,532,79	1,578,77
Gross Operating Revenues	1,210,00	1,246,30	1,283,68	1,322,20	1,361,86	1,402,72	1,444,80	1,488,14	1,532,79	1,578,77
Operating Expenses	725,395)	747,157)	769,572)	792,659)	816,438)	840,932)	866,160)	892,144)	918,909)	946,476)
Gross Operating Profit	484,605	499,143	514,117	529,541	545,427	561,790	578,644	596,003	613,883	632,300
<i>Gross Profit Percentage</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>	<i>%40</i>
Salaries and Benefits (Indirect Staff)	(77,440)	(80,615)	(83,920)	(87,361)	(90,943)	(94,671)	(98,553)	102,594)	106,800)	111,179)
General and Administrative Expenses	(17,396)	(17,918)	(18,456)	(19,009)	(19,580)	(20,167)	(20,772)	(21,395)	(22,037)	(22,698)
Marketing Expenses	(3,025)	(3,116)	(3,209)	(3,305)	(3,405)	(3,507)	(3,612)	(3,720)	(3,832)	(3,947)
Pre Operating Expenses	(2,750)									
Gross Indirect Expenses	100,611)	101,649)	105,585)	109,676)	113,927)	118,345)	122,937)	127,709)	132,669)	137,824)
Income before Interest, Depreciation and	383,994	397,494	408,532	419,865	431,500	443,445	455,707	468,294	481,214	494,476

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Fixed Assets Depreciations	(74,690)	(75,224)	(75,768)	(76,325)	(76,894)	(77,475)	(78,069)	(78,676)	(79,296)	(79,929)
Income before Tax and Interests	309,304	322,271	332,764	343,540	354,606	365,970	377,638	389,618	401,919	414,547
Bank Interests	(58,270)	(48,627)	(38,068)	(26,509)	(13,855)	0	0	0	0	0
Income Before Tax	251,033	273,644	294,695	317,031	340,752	365,970	377,638	389,618	401,919	414,547
Income Tax	(75,207)	(79,729)	(83,939)	(88,406)	(93,150)	(98,194)	100,528)	102,924)	105,384)	107,909)
Net Profit	175,827	193,915	210,756	228,624	247,601	267,776	277,110	286,695	296,535	306,637
<i>Net Profit Percentage</i>	<i>%15</i>	<i>%16</i>	<i>%16</i>	<i>%17</i>	<i>%18</i>	<i>%19</i>	<i>%19</i>	<i>%19</i>	<i>%19</i>	<i>%19</i>
Compulsory Reserves	(17,583)	(19,392)	(21,076)	(22,862)	(24,760)	(26,778)	(27,711)	(28,669)	(29,653)	(30,664)
Retained Earnings	158,244	332,768	522,448	728,210	951,052	1,192,05	1,441,44	1,699,47	1,966,35	2,242,32

Expected Cashflows Statement

The table bellow summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 41:Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	175,827	193,915	210,756	228,624	247,601	267,776	277,110	286,695	296,535	306,637
Bank Interests	58,270	48,627	38,068	26,509	13,855	0	0	0	0	0

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Depreciation	74,690	75,224	75,768	76,325	76,894	77,475	78,069	78,676	79,296	79,929
Total Operating Cashflows before Additions to Working Capital	308,787	317,765	324,593	331,459	338,350	345,251	355,179	365,370	375,830	386,566
Inventory (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Payables (Increase/Decrease)	20,066	602	620	639	658	678	698	719	740	763
Working Capital (Increase/Decrease)	(30,351)	(911)	(938)	(966)	(995)	(1,025)	(1,056)	(1,087)	(1,120)	(1,153)
Net Cashflows from Operating Activities	278,436	316,855	323,655	330,493	337,355	344,226	354,124	364,283	374,711	385,413
Cashflows from Investments Activities										
Fixed Assets (Procurement)	795,300)	(5,335)	(5,450)	(5,568)	(5,688)	(5,811)	(5,938)	(6,067)	(6,199)	(6,335)
	(									
Net Cashflows from Investments Activities	795,300)	(5,335)	(5,450)	(5,568)	(5,688)	(5,811)	(5,938)	(6,067)	(6,199)	(6,335)
	(									
Cashflows from Financing Activities										
Capital	409,778									
Loan Amortization	101,729)	111,373)	121,931)	133,490)	146,145)					
	(	(	(	(	(	0	0	0	0	0
Bank Interest Rate	(58,270)	(48,627)	(38,068)	(26,509)	(13,855)	0	0	0	0	0
Loans	614,667	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	864,446	159,999)	159,999)	159,999)	159,999)					
		(	(	(	(	0	0	0	0	0

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net (Increase/Decrease) in Cash	347,582	151,521	158,206	164,926	171,667	338,415	348,186	358,216	368,511	379,078
Cashflows at the Beginning of Period	0	347,582	499,103	657,309	822,235	993,903	1,332,317	1,680,503	2,038,720	2,407,231
Cashflows at the End of Period	347,582	499,103	657,309	822,235	993,903	1,332,317	1,680,503	2,038,720	2,407,231	2,786,309

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 42: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	347,582	499,103	657,309	822,235	993,903	1,332,317	1,680,503	2,038,720	2,407,231	2,786,309
Inventory	50,417	51,929	53,487	55,092	56,744	58,447	60,200	62,006	63,866	65,782
Total current Assets	0	0	0	0	0	0	0	0	0	0
Non Current Assets	397,999	551,032	710,796	877,327	1,050,647	1,390,764	1,740,704	2,100,726	2,471,097	2,852,091

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Fixed Assets (net)	720,610	650,722	580,403	509,645	438,439	366,775	294,644	222,035	148,939	75,345
Total Non Current Assets	720,610	650,722	580,403	509,645	438,439	366,775	294,644	222,035	148,939	75,345
Total Assets	1,118,609	1,201,754	1,291,199	1,386,972	1,489,086	1,757,540	2,035,348	2,322,761	2,620,036	2,927,436
Liabilities										
Current Liabilities	20,066	20,668	21,288	21,926	22,584	23,262	23,960	24,678	25,419	26,181
Remaining amount of Loan	111,373	121,931	133,490	146,145	0	0	0	0	0	0
Total current Liabilities	131,439	142,599	154,778	168,071	22,584	23,262	23,960	24,678	25,419	26,181
Non Current Liabilities										
Long Terms Loans	401,566	279,635	146,145	0	0	0	0	0	0	0
Total Long Term Liabilities	401,566	279,635	146,145	0	0	0	0	0	0	0
Total Liabilities	533,004	422,233	300,923	168,071	22,584	23,262	23,960	24,678	25,419	26,181
Owners Equity										
Shareholders Contributions	409,778	409,778	409,778	409,778	409,778	409,778	409,778	409,778	409,778	409,778
Statutory Reseve	17,583	36,974	58,050	80,912	105,672	132,450	160,161	188,830	218,484	249,148

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Retained Profits	158,244	332,768	522,448	728,210	951,052	1,192,050	1,441,449	1,699,474	1,966,355	2,242,329
Total Equity	585,605	779,520	990,277	1,218,901	1,466,502	1,734,278	2,011,388	2,298,083	2,594,617	2,901,255
Total Liabilities and Equity	1,118,609	1,201,754	1,291,199	1,386,972	1,489,086	1,757,540	2,035,348	2,322,761	2,620,036	2,927,436

Feasibility Indicators

There is more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 43:Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	(1,024,445)	281,186	311,520	318,205	324,925	331,667	338,415	348,186	358,216	368,511	379,078	5,837,912
Discount Factor	1.00	0.74	0.66	0.60	0.54	0.49	0.44	0.40	0.36	0.33	0.29	0.29
Net Present Value for Cash Flows	(1,024,445)	206,979	207,044	190,954	176,055	162,260	149,486	138,870	128,999	119,822	111,290	1,713,902

Summary of Feasibility Indicators

1. Payback Period

Table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 44: Payback Period

Payback	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(1,024,445)	281,186	311,520	318,205	324,925	331,667	338,415	348,186	358,216	368,511	379,078
Project Value	1,024,445	743,259	431,739	113,534	(211,392)	(543,058)	(881,473)	(1,229,659)	(1,587,875)	(1,956,386)	(2,335,465)
Payback Period	3	1	1	1	0	0	0	0	0	0	0

Table 45: Financial Analysis Result

Description	Value
Weighted Average Cost of Capital (WACC)	%10.75
Net Present Value for Cashflows	2,281,213
Payback Period	3
Internal Rate of Return	%36.19

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes the declined in revenues, changes in total investment costs, and increase in operating costs with 10%, which indicate that the project is feasible.

The following table shows that the project will be feasible if the revenues declined with 10% or the operating costs increase by 10%.

Table 46: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%36.19	3	10.75%
Revenues declined by 10%	%33.02	3	10.75%
Operating Expenses Increased by 10%	%30.51	4	10.75%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 2,281,213 Jordan Dinars considering that the project provides 11 Job Opportunities for the governorate residents.